

Volume 1 – Product Requirements Document (PRD)

AI Quant Trading Platform

Version: 1.0

Status: Draft

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1. Executive Summary

Product Name (Working Title)

TradeMind AI

Alternative Names

- QuantPilot AI
 - AlphaTrade AI
 - Nexus Quant
 - Quantum Trader AI
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2. Vision

Build the world's most intelligent AI-powered cryptocurrency trading platform that continuously analyzes the entire crypto market, identifies high-probability trading opportunities, manages risk automatically, and executes trades with minimal human intervention.

The platform should behave like an experienced quantitative trader, operating 24/7 using data-driven decisions rather than emotions.

3. Mission

Create a professional trading ecosystem where users do not need to monitor charts all day.

The platform continuously scans the market, evaluates opportunities, calculates probabilities, applies strict risk management, and only executes trades that satisfy predefined confidence thresholds.

4. Problem Statement

Retail traders face several challenges:

- Emotional decision making
- Fear of missing out (FOMO)
- Panic selling
- Lack of proper risk management
- Watching hundreds of coins manually
- Difficulty interpreting technical indicators
- Delayed reactions to market events
- No systematic trading process
- Inconsistent execution

Professional trading requires continuous monitoring and disciplined execution, which most individuals cannot maintain manually.

5. Proposed Solution

Develop an AI-powered automated trading platform capable of:

- Scanning hundreds of cryptocurrencies continuously
- Evaluating multiple technical indicators
- Analyzing market structure
- Monitoring sentiment and news
- Calculating trade probability scores
- Applying risk management automatically
- Executing trades through supported exchange APIs
- Monitoring open positions
- Generating reports and analytics

The system is designed to assist decision-making and automate predefined strategies. It does not guarantee profits or eliminate trading risk.

6. Product Objectives

Primary Objectives

- Reduce unnecessary losing trades
- Improve trade quality
- Automate repetitive trading tasks
- Remove emotional decision making

- Improve long-term consistency

Secondary Objectives

- Portfolio analytics
- Multi-exchange support
- AI-assisted trade recommendations
- Paper trading
- Strategy optimization
- Performance analytics

Business Objectives

- SaaS subscription model
 - Enterprise scalability
 - API-first architecture
 - Multi-tenant platform
 - White-label capability
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7. Target Audience

Primary Users

- Crypto traders
- Futures traders
- Swing traders
- Scalpers
- Portfolio managers

Secondary Users

- Trading communities
- Crypto influencers
- Investment groups
- Professional trading firms

Enterprise Users

- Fund managers
 - Proprietary trading firms
 - Quantitative trading teams
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8. User Personas

Persona 1 – Beginner

Needs:

- Automated assistance
- Easy dashboard
- Risk control
- Learning insights

Persona 2 – Intermediate Trader

Needs:

- Custom strategies
- Risk management
- Notifications
- Analytics

Persona 3 – Professional Trader

Needs:

- API integration
- Strategy customization
- Portfolio management
- Multi-exchange trading

Persona 4 – Institutional User

Needs:

- High availability
- Team management
- Audit logs
- Advanced reporting

9. Core Value Proposition

Instead of reacting to the market, the platform continuously observes, evaluates, filters, and acts only on predefined high-confidence conditions.

The objective is disciplined execution based on data and risk controls.

10. Key Product Principles

- Data-driven decisions
- Risk-first approach
- Automation over manual repetition
- Transparency

- Explainable AI decisions
 - User control over automation
 - High reliability
 - Security by design
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11. Scope

Included

- Market scanning
- Technical analysis
- AI scoring
- Strategy execution
- Risk management
- Portfolio management
- Notifications
- Reporting
- Exchange integrations
- Admin panel

Not Included (Version 1)

- Stock market
 - Forex
 - Commodity trading
 - Copy trading marketplace
 - Options analytics
 - High-frequency trading
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12. User Journey

Step 1 Create account



Step 2 Verify email



Step 3 Enable 2FA



Step 4 Connect exchange API



Step 5 Select strategy



Step 6 Configure risk profile



Step 7 Paper trading (recommended)



Step 8 Review performance



Step 9 Enable live trading



Step 10 Monitor portfolio and reports

13. Functional Requirements

The platform shall:

- Allow user registration
 - Support secure authentication
 - Store encrypted API credentials
 - Connect supported exchanges
 - Continuously scan markets
 - Generate AI probability scores
 - Execute trades based on configured rules
 - Manage stop loss and take profit
 - Generate reports
 - Send notifications
 - Maintain audit logs
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14. Non-Functional Requirements

Availability

99.9%

Performance

Real-time updates

Scalability

Support thousands of users

Security

Encryption

JWT Authentication

RBAC

API Rate Limiting

Monitoring

Centralized logging

Alerting

Audit trails

Reliability

Automatic recovery

Backup

Disaster recovery

15. Success Metrics

Product Metrics

- Active users
- Daily active traders
- Strategy adoption
- User retention
- Paper-to-live conversion

Operational Metrics

- API latency
- Scanner speed
- Signal generation time

- System uptime
- Error rate

Business Metrics

- Monthly recurring revenue
 - Customer lifetime value
 - Churn rate
 - Subscription conversion
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16. Monetization

Free Plan

- Paper trading
- Limited signals
- Basic dashboard

Starter Plan

- Live trading
- One exchange
- Standard strategies

Professional Plan

- Multiple exchanges
- AI scoring
- Advanced analytics

Enterprise Plan

- Unlimited users
 - Team management
 - White-label
 - Dedicated infrastructure
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17. Risks

Business Risks

- Exchange API changes
- Regulatory changes
- Competition

Technical Risks

- API downtime

- Data delays
- Incorrect market data
- Network failures

Trading Risks

- Market volatility
- Slippage
- Liquidity constraints
- Unexpected events

The platform must clearly communicate that trading involves financial risk and cannot guarantee returns.

18. Product Roadmap

Phase 1

- Authentication
- Dashboard
- Exchange integration
- Portfolio
- Paper trading

Phase 2

- Technical analysis
- Risk engine
- Automated execution

Phase 3

- AI scoring
- Sentiment analysis
- Advanced reporting

Phase 4

- Machine learning
- Strategy optimization
- Multi-exchange automation

Phase 5

- Enterprise edition
 - White-label
 - Institutional features
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19. Future Vision

Future versions may include:

- AI-generated trading strategies
 - Natural language trading assistant
 - Cross-exchange arbitrage detection
 - On-chain analytics
 - Options support
 - Institutional compliance tools
 - Mobile applications
 - Multi-language support
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20. Disclaimer

This platform is intended to assist users through automation, analytics, and systematic execution. Cryptocurrency trading involves substantial financial risk. The platform is designed to improve decision quality through data analysis and risk management but does not guarantee profits or prevent losses.